

Meningitis beyond the neonatal age

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Central Nervous System Infections

- Significant cause of morbidity and mortality in children.
- Classified according to the affected tissue:-
 - Meningitis:-
 - Encephalitis indicates brain parenchymal involvement.
- ❖ These anatomic boundaries may be indistinct during infection.

Bacterial meningitis

DEFINITION

-Acute & diffuse bacterial infection of the CNS
with primary involvement of the meninges .

ETIOLOGY

➤ In general: *viruses* > *bacteria* > *fungi* > *parasites*, in terms of ...

- Pathogen is influenced by :

- ✓ Age of the host
- ✓ Immune status of the host
- ✓ Epidemiology of the pathogen

a) Birth - 2 months

- ✓ Group B Streptococci (*Streptococcus agalactiae*)
- ✓ Gram –negative enteric bacilli(*E.coli*, *klebsiella*)
- ✓ *Listeria Monocytogenes*
- ✓ Group D streptococci (*Enterococcus*)

b) 2 month - 12 years

- ✓ *S. Pneumoniae*
- ✓ *N. Meningitidis*
- ✓ H.. influenza type B (70% of cases of meningitis in **<5 yr of age**)

c) Other less common pathogens

- ✓ *P. aeruginosa*, *S. aureus*, CONS, *Salmonella* spps, *L. monocytogenes*

Etiology..

➤ H. influenzae

- Invasive infections occur primarily in the 1st 2 month -2 yrs of life
- Peak incidence 6 - 9 month of age
- Risk factors :
 - ✓ Family or day care center contacts of patients with H . inf type b disease
 - ✓ Unvaccinated individuals
 - ✓ Individuals with blunted immunologic response to vaccine
e.g. child with HIV infection

➤ S. Pneumoniae

- Invasive infection peaks during the first 2 yrs of life
- Risk factors :
 - ✓ Age < 2 yrs
 - ✓ Asplenia (functional or anatomic)
 - ✓ HIV infection
 - ✓ Otitis media, sinusitis, pneumonia
 - ✓ CSF otorrhea or rhinorrhea
 - ✓ Presence of cochlear implant: risk of meningitis increased by more than **30-fold**

Etiology...

➤ N. meningitidis

- ✓ Serogroups:- A ,B ,C, X, W135,Y
 - Epidemic disease **serogroup A**
- ✓ Cases are common in winter & spring: ...
- ✓ Nasopharyngeal carriage:- 1-15 % of adults
- ✓ Attack rate in family.....1 %
- ✓ **Bimodal peak:- younger than 5 yr** and b/n **15-24 yr** of age
- ✓ Most infections of children are acquired from :
 - Contact in a day care facility
 - Colonized adult family member
 - Ill patient with meningococcal disease

General risk factors

- ✓ **Lack of immunity** associated with young age.
- ✓ **Male gender**
- ✓ **Black race**
- ✓ **Infants & young children with occult bacteremia:-**the odds ratio is greater for meningococcus (85 times) and H. influenzae type b (12 times) relative to that for pneumococcus.

Specific risk factors...

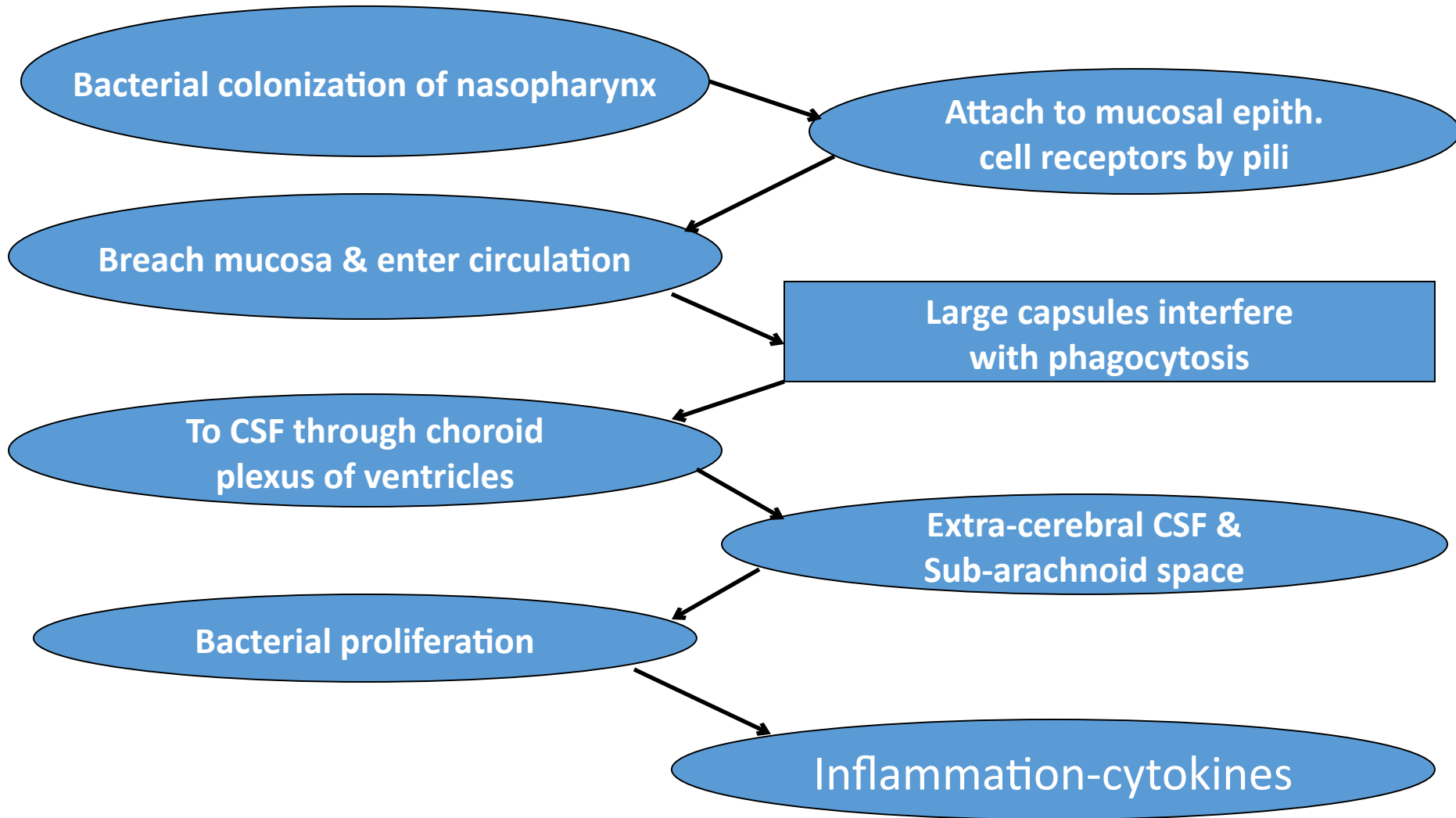
- **Complement system** (C5-C8), **properdin defects** (N meningitidis)
- **T-lymphocyte defects** (L. monocytogenes)
- **Congenital or acquired CSF leak** (pneumococcus)
- Lumbosacral dermal sinus and **meningomyelocele**
- ✓ (staphylococcal & gram-negative enteric bacteria)
- **CSF shunt infections** (staphylococci spp,.....)

Epidemiology

- ✓ **Mode of transmission:-** Person to person contact through **respiratory tract secretions or droplets**
- ✓ **Meningococcal meningitis** is the **only** form of bacterial meningitis **that causes outbreaks.**
- ✓ The largest outbreaks occur mainly in the semi-arid areas of sub-Saharan Africa, designated the **African “meningitis belt”.**
- ✓ Epidemics previously occurred every 8-12 yrs, however in recent years they have been occurring yearly.
- ✓ **Serogroup A** accounts for **80–85%** of out-breaks.



Pathogenesis: Out side the neonatal period is typically due to bacterial colonization of the nasopharynx with subsequent invasion into the bloodstream,



Pathogenesis...

- Rarely, meningitis may follow **bacterial invasion** from a **contiguous focus** of infection such as:-
 - ✓ Paranasal sinusitis,
 - ✓ Otitis media,
 - ✓ Mastoiditis,
 - ✓ Orbital cellulitis, or cranial or vertebral osteomyelitis
 - ✓ Via penetrating cranial trauma, dermal sinus tracts, or meningocele

➤ Pathology and pathophysiology

- **Meningeal exudates** (with variable thickness)
- Ventriculitis:
- Subdural effusions
 - ✓ Occurs in **later phase**:- because of **transudation of fluid**
- Subdural empyema
- **Perivascular inflammatory infiltrates**



Pathology and pathophysiology ...

- **Cerebral infarction:** May range from microscopic to an entire hemisphere.
- **Inflammation of the spinal nerves and roots**
 - ✓ Produces meningeal signs
- **Inflammation of the cranial nerves**
 - ✓ Produces cranial neuropathies(2,3,7,8)
- **Increased ICP**
 - ✓ Uncal herniation.....temporal lobe compression of oculomotor nerve...3 nerve palsy
 - ✓ **Types of cerebral edema :**
 - i. Cell death (**cytotoxic** cerebral edema)
 - ii. Cytokine-induced increased capillary permeability (**vasogenic** cerebral edema)
 - iii. Increased hydrostatic pressure (**Interstitial** cerebral edema)
- **SIADH**
 - ✓ Excessive water retention → Increased risk of elevated ICP
 - ✓ Hypotonicity of brain extra cellular spaces → cell swelling and lysis → Cytotoxic edema

CLINICAL MANIFESTATIONS

➤ The severity and constellation of signs are determined by host–pathogen interactions and the affected region of the CNS

➤ **Two patterns of onset :**

a) Dramatic onset (less common)

✓ Shock, purpura, DIC, reduced LOC, death within 24 hours.

b) Gradual onset (more common)

✓ Meningitis is preceded by several days of fever accompanied by URT or GI symptoms:

-followed by nonspecific signs of CNS infection like increased lethargy & irritability .



CLINICAL MANIFESTATIONS...

- Non-specific findings:
 - ✓ Fever, anorexia, poor feeding, symptoms of URTI, myalgias, arthralgias, ↑HR, ↓BP, petechiae, purpura, erythematous macular rash
- **Signs of meningeal irritation** :- Back pain, neck stiffness, Kernig sign, Brudzinski sign.
- Papilledema, photophobia
- Focal neurologic signs:- 10 - 20% of cases
- Cranial neuropathies...

CLINICAL MANIFESTATIONS...

➤ Signs of increased ICP

- Headache, emesis, bulging fontanel, diastasis (widening) of the sutures
- Oculomotor or abducens nerve palsy
- HTN with bradycardia; stupor, coma
- Apnea or hyper ventilation, signs of herniation
- Decorticate or decerebrate posturing

➤ Seizures

- Focal or generalized
- 20 -30 % of cases
- Causes : cerebritis, infarction, or electrolyte disturbance

➤ Alteration of mental status

- Common
- Causes : increased ICP, cerebritis, hypotension
- Manifestations : irritability, lethargy, stupor, coma.

Skin lesions in acute meningococccemia can begin as papules but quickly progress to petechiae & purpura. As seen here, the purpuric lesions can coalesce



Diagnosis

- ✓ CSF analysis and culture
- ✓ Blood cultures (positive in 80% to 90% of cases)

** When you need to evaluate complications*

- ✓ Neuro-imaging (Brain U/S, CT, MRI)
- ✓ Urinalysis and serum sodium

Lumbar puncture



Lumbar puncture

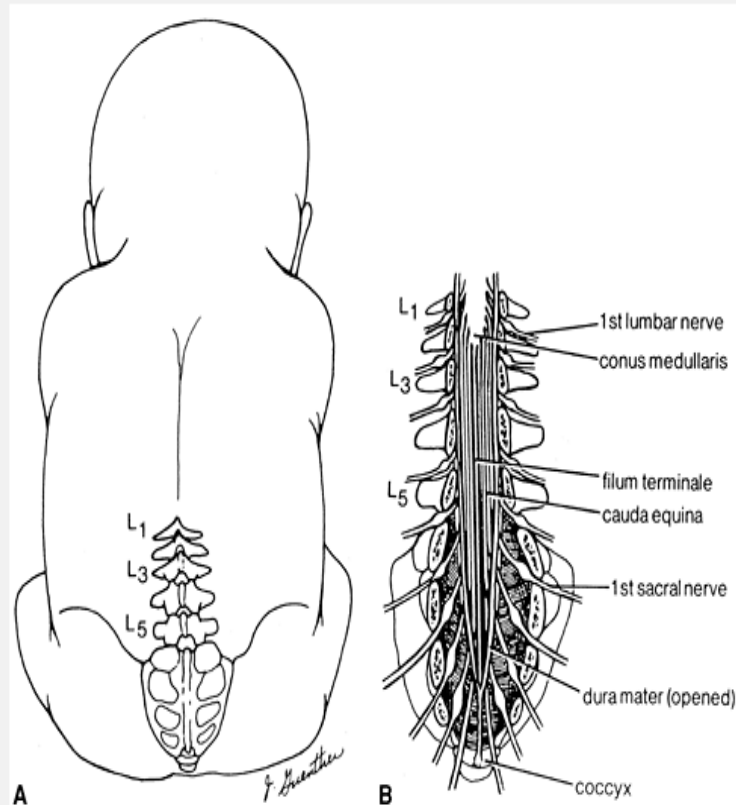


FIG. 16.3. A: Externally palpable anatomic landmarks. **B:** Vertebral bodies removed to show anatomy of spinal cord in lumbosacral area in relation to external landmarks.

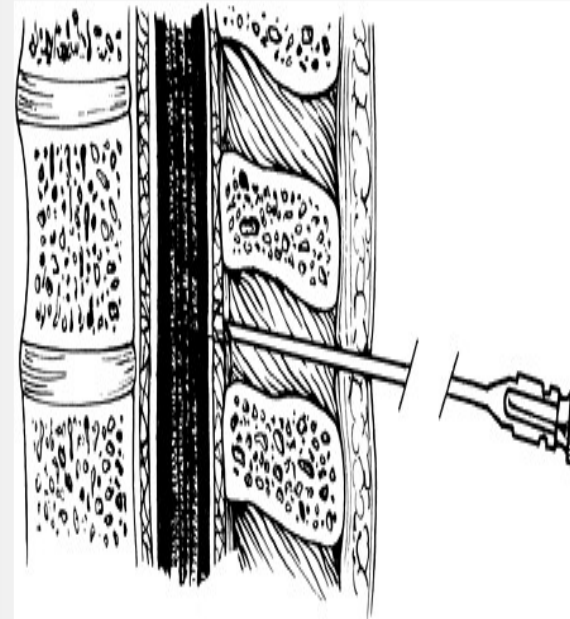


FIG. 16.5. Needle has penetrated the dura, and stylet has been removed to allow free flow of spinal fluid.

Lumbar puncture

INDICATIONS:-...

- ✓ **To diagnose** central nervous system (CNS) infections
- ✓ To monitor efficacy of antimicrobial therapy in CNS infection
- ✓ To drain CSF in communicating hydrocephalus associated with IVH
- ✓ **To diagnose** intracranial hemorrhage
- ✓ **To diagnose** CNS involvement with leukemia
- ✓ To inject chemotherapeutic agents
- ✓ To instill contrast material for myelography **7**

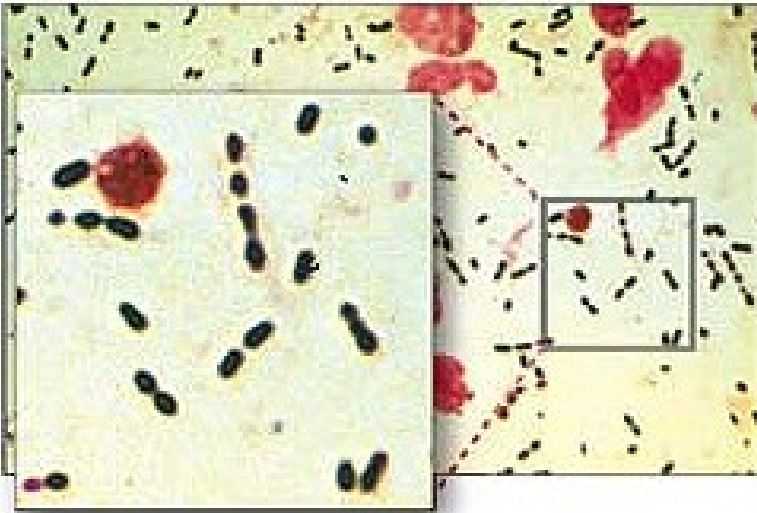
Lumbar puncture

➤ **CONTRAINDICATIONS:**

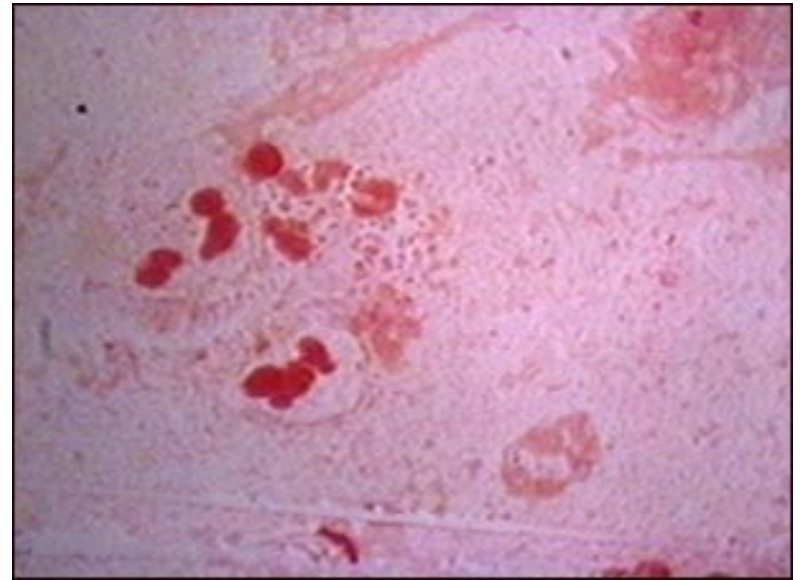
- ✓ Increased intracranial pressure (ICP)
- ✓ Uncorrected thrombocytopenia or bleeding diathesis
- ✓ Infection in the skin or underlying tissue at or near the puncture site
- ✓ Lumbosacral anomalies
- ✓ Cardiorespiratory instability, which may be exacerbated by the procedure

CSF analysis

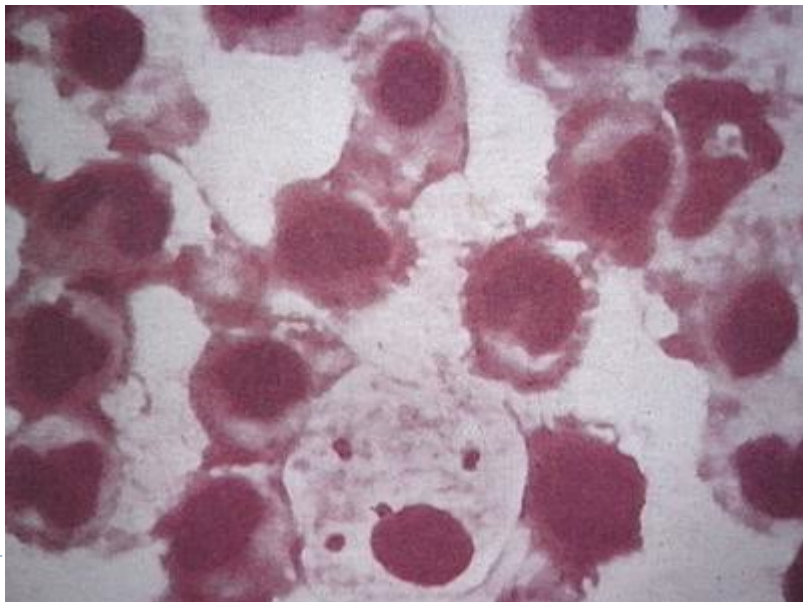
- In bacterial meningitis:- CSF shows ↑ pressure, ↑ protein (BBB inflammation), ↓ glucose (altered glucose transport by the cerebral tissue...), and many neutrophils.
 - * turbid/frank pus – a minimum of 200 cells/mm³
- ✓ Presence of infecting organism/antigenic capsular substance confirms dx
- Gram stain is positive **in > 70%** of patients with untreated bacterial meningitis..
- **Traumatic** LP may also complicate the interpretation of CSF tests:



S Pneumoniae



H influenzae



N meningitidis

CSF analysis...

- Cultures often are negative if the patient has been treated with antibiotics prior to culture.
- ✓ Detection of capsular polysaccharide antigens by **rapid latex agglutination tests** in CSF can support **diagnosis in such cases.**

DIFFERENTIAL DIAGNOSIS

A. INFECTIONS

1. Generalized infection of the CNS

✓Bacteria

- M. Tuberculosis (Tb meningitis)
- T. Pallidum (Syphilis)

✓Fungi

- Histoplasma ,Candida ,Cryptococcus Aspergillus

✓Parasites

- T. gondii , Cysticercosis, Malaria

✓Viruses

- Enteroviruses, HSV (Viral meningoencephalitis)

DDX...

2. Focal infections of the CNS

- **Brain abscess**
- **Para meningeal abscess**
 - Subdural empyema
 - Cranial epidural empyema
 - Spinal epidural empyema

B. NON-INFECTIOUS ILLNESSES

- Cause generalized inflammation of the CNS
- Uncommon
 - **Malignancy**
 - **Collagen vascular syndromes**
 - **Exposure to toxins**
- ❖ **All acute febrile illnesses** can be D.DX of acute bacterial meningitis especially in its early phase.

Form of meningitis	Opening pressure	WBCs & differential	Protein	Glucose	Comments
Normal values	< 28 cm H ₂ O	< 5, >75% lymphocytes	20 - 45 mg/dl	> 50 mg/dl or 75% of ..	
Acute bacterial meningitis	Usually ↑ (100 to 300)	Usually 300 – 2,000; PMNs predominate	Usually 100 – 500	Usually < 40	Organisms usually seen on Gram stain / culture
Partially treated bacterial meningitis	Normal or elevated	5 – 10,000; PMNs usual	Usually 100 – 500	Normal or decreased	
Viral meningitis	Normal or slightly ↑ (80 – 150)	Rarely >1,000 cells. Mono-nuclear cells predominate	Usually 50 – 200	Generally normal	
TB meningitis	Usually elevated	10 – 500; lymphocytes predominate	100 – 3,000	< 50 in most cases	Acid-fast organisms rarely seen on smear

Principles of treatment

- ✓ It is a medical emergency
- ✓ Always use high dose, parenteral (IV) antibiotics as soon as the LP is done
- ✓ IV antibiotics as soon as possible if there are contraindications for LP; then do a CT scan
- ✓ Treat raised ICP and seizures (if present)
- ✓ Supportive care: feeding, Follow with..
- ✓ IV dexamethasone QID for 2 days (> than 6 wks of age:- give 1-2 hr before antibiotics are initiated or concurrently with the 1st dose of antibiotics.

Treatment...

- Empirical antibiotic regimen in a suspected case of meningitis outside the neonatal period is:
 - ✓ **Vancomycin combined with a third generation cephalosporin (ceftriaxone).**
 - ✓ Can be treated with penicillin G or ampicillin
 - ✓ Chloramphenicol can also be added to penicillin.
 - Even though it's a bacteriostatic antibiotic, it is bactericidal for *S. Pneumoniae*, *H. influenzae*, and *N. meningitidis*

Treatment...

❖ Duration of treatment

➤ Generally total of 10 days

➤ *Specific (based on etiologic agent) in uncomplicated cases*

- | | |
|-------------------|------------|
| ✓ N meningitidis | 5-7 days |
| ✓ H influenzae | 7-10 days |
| ✓ S pneumoniae | 10-14 days |
| ✓ L monocytogenes | 14-21 days |
| ✓ Enteric GN rods | 3 wks |

Acute complications

- ✓ Seizures
- ✓ Raised ICP
- ✓ DIC, shock
- ✓ Cranial nerve palsies
- ✓ Stroke
- ✓ Cerebral or cerebellar herniations
- ✓ Thrombosis of the dural venous sinuses
- ✓ Subdural effusion
- ✓ Ventriculitis with hydrocephalus
- ✓ SIADH

Chronic complications

- ✓ Visual impairment
- ✓ Sensorineural hearing loss
- ✓ Behavioral problems
- ✓ Cognitive dysfunction
- ✓ Recurrent seizures
- ✓ Delay in acquisition of language

PROGNOSIS

➤ POOR PROGNOSTIC FACTORS

- ✓ Pneumococcal meningitis
- ✓ Age < 6 months
- ✓ **>10⁶ colony** – forming units of bacteria / ml of CSF
- ✓ **Seizure** occurring after **4 days of therapy**
- ✓ **Coma or focal neurological** signs on presentation

➤ **Mortality:** < 10 % with antibiotic therapy and supportive care

- ✓ Severe neurodevelopmental sequelae:- 10 -20 % of cases
- ✓ Neurobehavioral morbidity:- 50 % of cases

➤ Most common neurologic sequelae

- ✓ Hearing loss
- ✓ Mental retardation , delay in acquisition of language
- ✓ Seizures
- ✓ Visual impairment , behavioral problems

Prevention

- Conjugate **pneumococcal vaccines** (Eg PCV10)
- Conjugate **Hemophilus influenzae type b vaccine**
- Conjugate **meningococcal vaccine** (MCV4)
 - ✓ Contains capsular polysaccharides from serogroups A, C, W-135, and Y conjugated to diphtheria toxoid.

Prophylaxis against meningococcal meningitis

- Recommended **for all close contacts of patients** with meningococcal meningitis, regardless of age or immunization status.
- Prophylaxis **not routinely recommended for medical personnel** except those with intimate exposure (mouth-to-mouth resuscitation, intubation, suctioning before antibiotic therapy was begun).
- ✓ Children may be given **Rifampin** or **Ceftriaxone**
- ✓ **Ciprofloxacin** (500 mg PO stat) may be given to persons ≥ 18 yr.
- ✓ Penicillin doesn't eradicate nasopharyngeal carriage; patients treated with penicillin should get prophylaxis before discharge.

Prophylaxis against *H. influenzae* type b meningitis

- **Rifampin** prophylaxis should be given
 - ✓ to **all household contacts** of patients with invasive disease caused by *H. influenzae* type b,
 - ✓ if any close family member **younger than 48 mo has not been fully immunized** or
 - ✓ if **an immunocompromised** person, of **any age**, resides in the household.

Tuberculous Meningitis

Etiology & Epidemiology

Etiology

- ✓ Mycobacterium Tuberculosis
- ✓ Non-spore forming, non-motile, pleomorphic ,weakly gram-positive, acid-fast bacilli

Epidemiology

- ✓ Occurs in 0.3% of untreated TB infections
- ✓ Peak age: *6 mo - 4 yr of age*
- ✓ **BCG vaccination** prevents it

Pathology and Pathophysiology

- Lymphohematogenous dissemination of the primary (lung in $> 90\%$ of cases) infection leads to formation of metastatic caseous lesion in the cerebra cortex or meninges
- Exudates formation
- **Salt wasting or SIADH** result in abnormal electrolyte metabolism
- Vasculitis, infarction, cerebral edema, and hydrocephalus
- Brain stem: site of greatest involvement

Clinical Manifestations

➤ Clinical course :

A. Rapid progression

- ✓ Infants & young children, occurs in *several days* before the onset of acute hydrocephalus, seizures, and cerebral edema.

B. Slow progression

- ✓ Occurs in several weeks
- ✓ Has 3 stages

➤ 1st Stage

- ✓ Lasts **1-2 weeks**
- ✓ Non specific symptoms:-
- ✓ Infants can experience a stagnation or **loss of developmental milestones**.

➤ 2nd Stage

- ✓ Abrupt in onset
- ✓ **Lethargy, +ve meningeal signs**, seizures, hypertonia, vomiting, cranial nerve palsy, **other focal neurologic signs**, signs of encephalitis with no meningeal signs

➤ 3rd Stage

- ✓ **Coma, hemiplegia or paraplegia**, hypertension
- ✓ Decerebrate posturing, deterioration of vital signs, and finally death.

Diagnosis

- PPD: -ve in 50% of cases
- CXR: Normal in 20-50% of cases
- CSF (Specimen volume: 5-10 ml)
 - ✓ WBC : 10-500/mm³ (Lym. predominant)
 - ✓ Glucose : <40mg/dl (rarely <20mg/dl)
 - ✓ Protein: 400-5000mg/dl
 - ✓ Smear for AFB: +ve up to 30% of cases
 - ✓ Culture: +ve in 50-70% of cases
 - ✓ PCR : may be positive

➤ CT or MRI of the brain

- ✓ Normal in early stages of TB meningitis
- ✓ Basilar enhancement, communicating hydrocephalus, signs of cerebral edema, early focal ischemia
- ✓ One or more clinically silent *tuberculomas* may be seen in the cerebral cortex or thalamic areas
- ❖ **N.B:** Dx of TB meningitis can be difficult early in its course and requires **HIS**

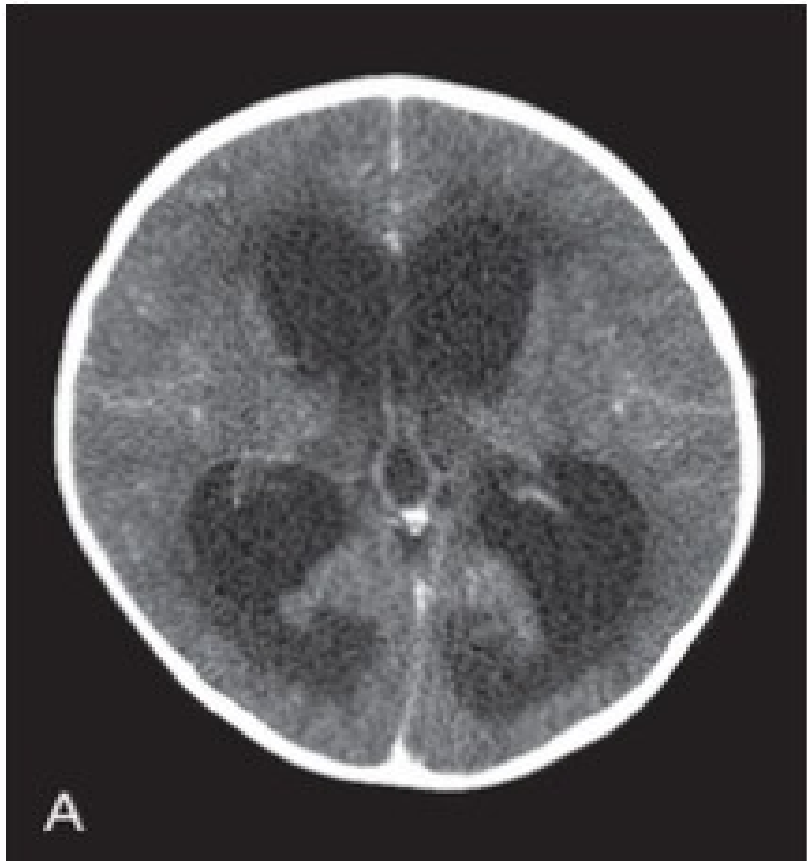


FIG. 242.17 MRI of brain of 3 yr old child showing multiple pontine tuberculomas.

Treatment

❖ Anti TB

➤ Directly Observed Therapy (DOTs Regimen)

✓ 2RHZ E or S / 7-10 RH: - (Total: 9-12 mo.)

❖ Steroids

✓ Prednisone, 1-2 mg/kg/day in 1-2 divided doses orally for 4-6 weeks , followed by gradual tapering

❖ Rx of complications:

✓ Increased ICP , Seizures, Hydrocephalus (CSF shunts)

➤ **Complications**

- ✓ Blindness, deafness, paraplegia or hemiplegia, mental retardation, DI, hydrocephalus, cranial nerve palsy, seizures

➤ **Prognosis**

- ✓ Correlates with clinical stage at time of initiation of Rx:
 - 1st stage has excellent outcome
 - 3rd stage survivors have permanent disabilities
- ✓ Better for older child than for young infant

➤ **Prevention**

- ✓ Prevention of primary Tb infection :
 - **Vaccination (BCG)**
- ✓ Early Dx & Rx of TB cases

Viral Meningoencephalitis

➤ Definition

- It is an acute inflammatory process involving the meninges and/or brain tissue to a **variable degree**.

➤ Etiology

- **Enteroviruses** (coxsackievirus, echovirus, enterovirus, poliovirus)
 - Most common cause, small RNA viruses
 - More than 80 serotypes
- **Herpes viruses**
 - *HSV-1*, HSV-2, **VZV**, CMV, EBV, HHV-6
- **Mumps, HIV**

Etiology ...

- **Arboviruses:** broad range of viruses from **multiple viral families**
 - Mosquitoes and ticks are most common vectors
 - Tend to affect the **entire brain**
 - Birds or small animals**Humans**
e.g. West Nile virus
- **Other viruses/occasional causes/**
 - Rubella, rubeola, rabies
 - Respiratory viruses : Adenovirus, Influenza virus, parainfluenza virus
- **Live virus vaccinations/rare causes/**
 - Polio, measles, mumps, rubella

➤ **Epidemiology**

✓ Enteroviruses :

- Direct spread from person to person, IP is 4-6 days

✓ Other factors : Season, geography, climate conditions, animal exposures, etc.

➤ **Pathogenesis and pathology**

- Neuronal disruption, tissue necrosis, meningeal congestion

- Neuronal damage from :

- 1) Direct invasion and destruction of neural tissues by virus

- 2) Host reaction to viral agents

Clinical Manifestations

- ✓ **Determinant factors are :**
 - 1) Relative degree of meningeal & parenchymal involvement
 - 2) Specific etiology
- ✓ **Mild self-limited** illness to severe encephalitis with sequelae or resulting in death
- ✓ **Fever**, headache, nausea, vomiting, photophobia, pain of (neck, back and legs).
- ✓ Nuchal rigidity, irritability, change in mental status, convulsions, bizarre movements, hallucinations
- ✓ **Exanthems precede** or accompany **CNS signs**

Diagnosis

➤ Mainly clinical :

- **Nonspecific prodrome** followed by progressive CNS symptoms

➤ CSF

- WBC : 10-100/mm³(up to 1000/mm³)

Mononuclear cells predominate later

- **Protein** : 50-200 mg/dl

- Glucose : normal (**mumps** <40mg/dl)

- **PCR** : for **HSV** & **enteroviruses**

- Culture : for enteroviruses(70% detection)

Diagnosis...

➤ Serologic tests:

- For arboviruses
- Not practical for enteroviruses

➤ **EEG**: diffuse slow-wave activity usually without focal changes

➤ **CT/MRI**: swelling of the brain parenchyma

❖ **N.B**: Focal seizures or focal findings on EEG, CT, or MRI, especially involving the **temporal lobes** suggest **HSV encephalitis**.

Treatment

1. Supportive

- Rest
- Non-aspirin containing analgesics: Acetaminophen is best
- Reduction in room light ,noise
- IV fluids : if poor oral intake
- In severe cases : ICU care

2. Acyclovir for HSV encephalitis

2. Rx of complications :

- Increased ICP, SIADH ,seizures, etc.

Complications

- Increased ICP(cerebral edema)
- Coma
- Convulsions
- Fluid & electrolyte imbalance
- Aspiration , Asphyxia
- SIADH
- Cardiac or respiratory arrest of central origin

Prognosis

- Most children recover completely
- **Depends on :**
 - ✓ Severity of clinical illness
 - ✓ Specific etiologic agent
 - ✓ Age of the child (Poor for infants)
- Potential deficits in severe cases :
 - Intellectual, motor, epileptic, psychiatric,
 - Auditory, visual

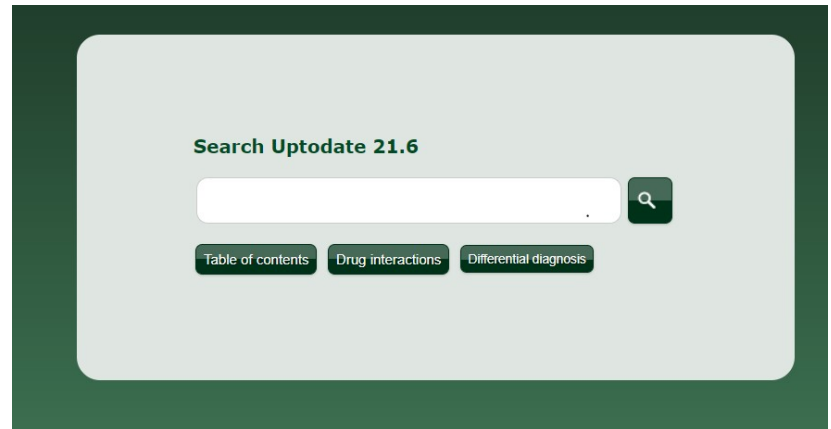
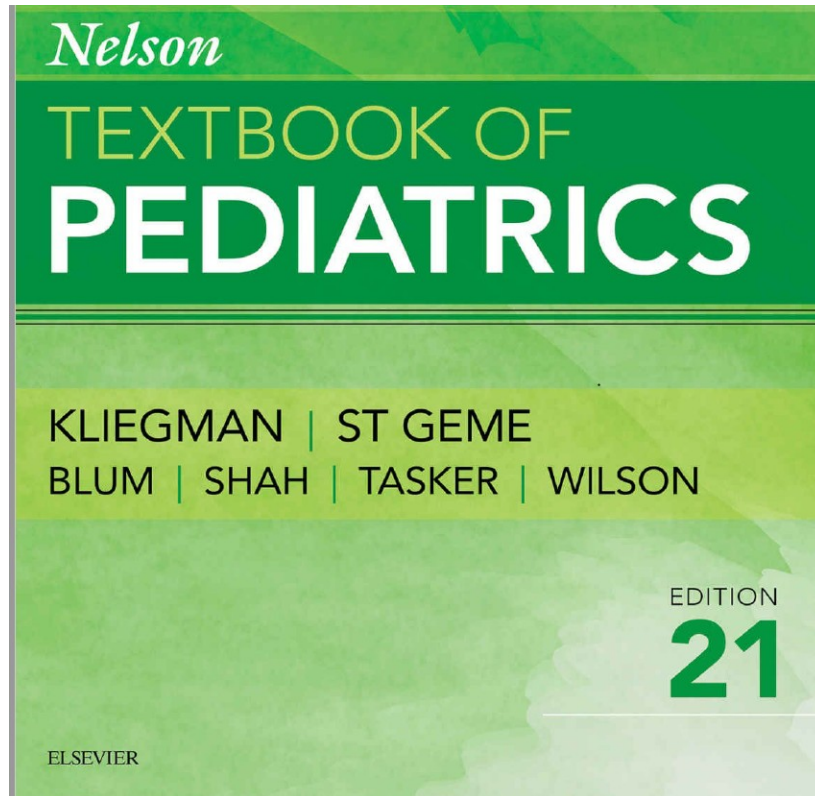
Prevention

➤ **Vaccination**

- **Polio**
- **Measles**
- **Mumps**
- **Rubella**
- **Rabies**

➤ Control of insect vectors & minimizing mosquito bites

REFERENCES



THANK YOU